

Lab 3: JavaScript Fundamentals and Frontend Basics

Course: CST150: Web technologies

Objective

By the end of this lab, students will:

- Understand JavaScript fundamentals.
- Learn how to manipulate the DOM.
- Implement interactive elements on a web page.

As always, don't just follow the workshop verbatim, tweak the changes and see the difference. Share your findings with the group.

Pre-requisites

- Completion of Lab 2 (HTML, CSS, and Responsive Design).
 - Basic knowledge of HTML and CSS.
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Lab Tasks

Task 1: Setting Up JavaScript in Your Project

1. Duplicate `CST150_lab2` folder into `CST150_lab3` folder.
2. Create a new file named `script.js`.
3. Link `script.js` to your `index.html` file by adding the following before the closing `</body>` tag:

```
<script src="script.js"></script>
```

Task 2: JavaScript Basics - Variables and Functions

1. Open `script.js` and add the following code:

```
// Variables and Data Types
let message = "Hello, JavaScript!";
console.log(message);
```

```
// Function Declaration
function greetUser() {
  alert("Welcome to JavaScript Fundamentals!");
}
```

2. Call the function when the page loads by adding this at the end of `script.js`:

```
document.addEventListener("DOMContentLoaded", greetUser);
```

Task 3: Interacting with the DOM

1. Modify `index.html` to include a button:

```
<button id="changeText">Click Me</button>
<p id="text">This text will change.</p>
```

2. Update `script.js` to add event handling:

```
// DOM Manipulation
let button = document.getElementById("changeText");
let text = document.getElementById("text");

button.addEventListener("click", function() {
  text.textContent = "You clicked the button!";
  text.style.color = "blue";
});
```

Task 4: Debugging and Console Logging

1. Open the browser developer console (F12 or right-click → Inspect → Console).
2. Observe the `console.log(message)` output.
3. Click the button and watch for changes in the console.

Discussion Questions

1. How does JavaScript enhance frontend interactivity?
2. What is the role of the `DOMContentLoaded` event?
3. How would you modify the button to toggle text color between red and blue?

Bonus Challenge

- Create an input field where users can enter text that updates the paragraph when they type.
- Use `setInterval()` to create a simple real-time clock on the page.

Conclusion

In this lab, students learned the basics of JavaScript, including variables, functions, and DOM manipulation. Future lessons will focus on event-driven programming and integrating JavaScript with backend systems.

End of Lab 3